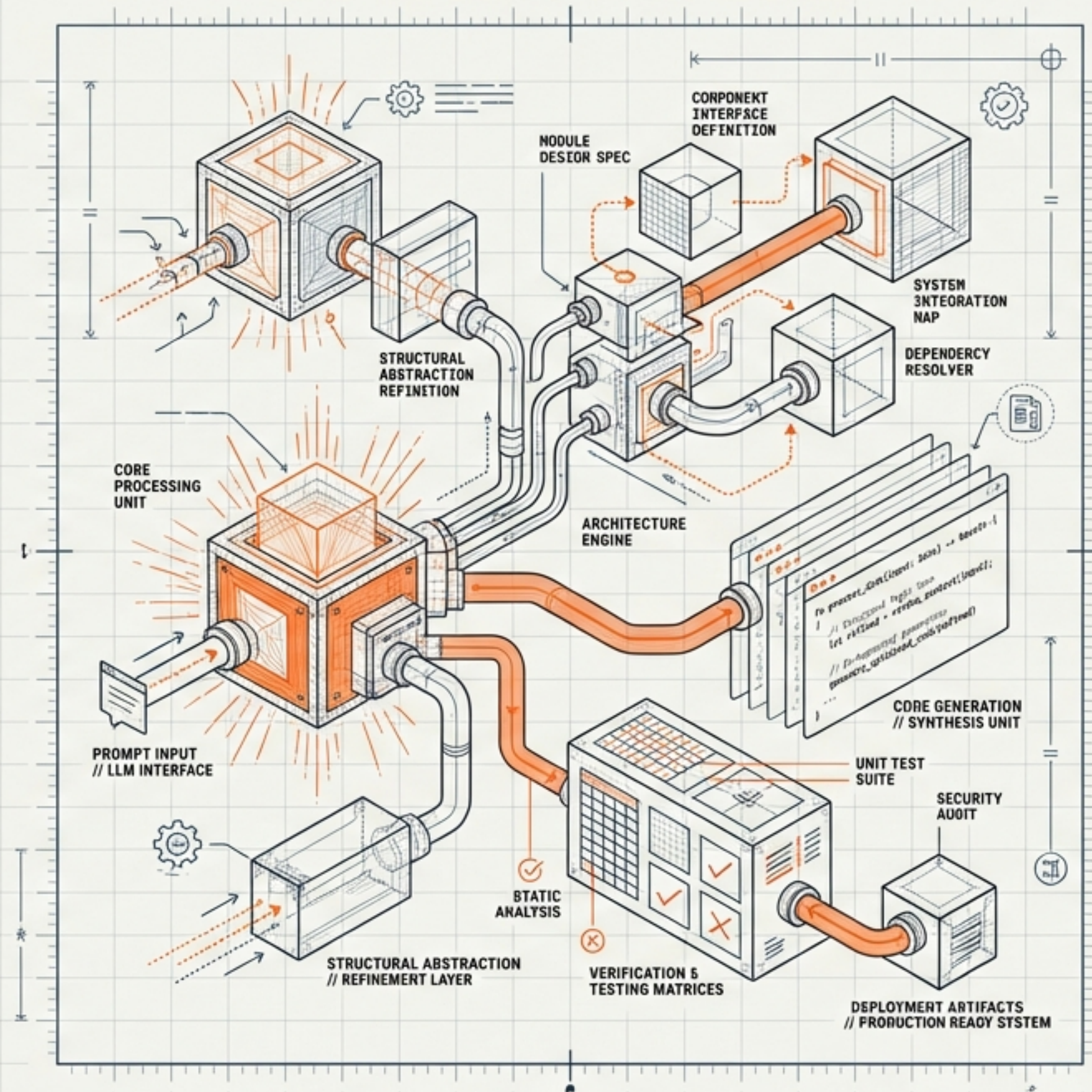
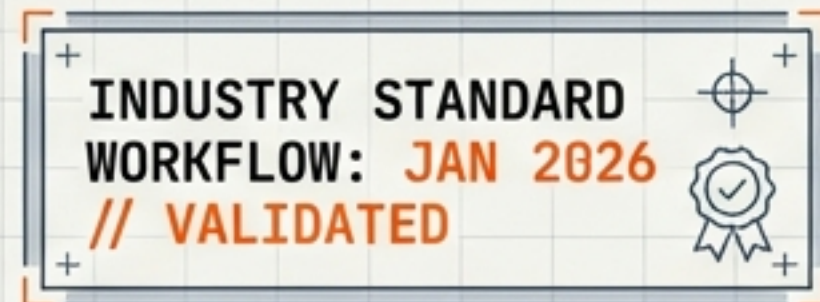


The AI-Augmented Developer

A Disciplined Workflow

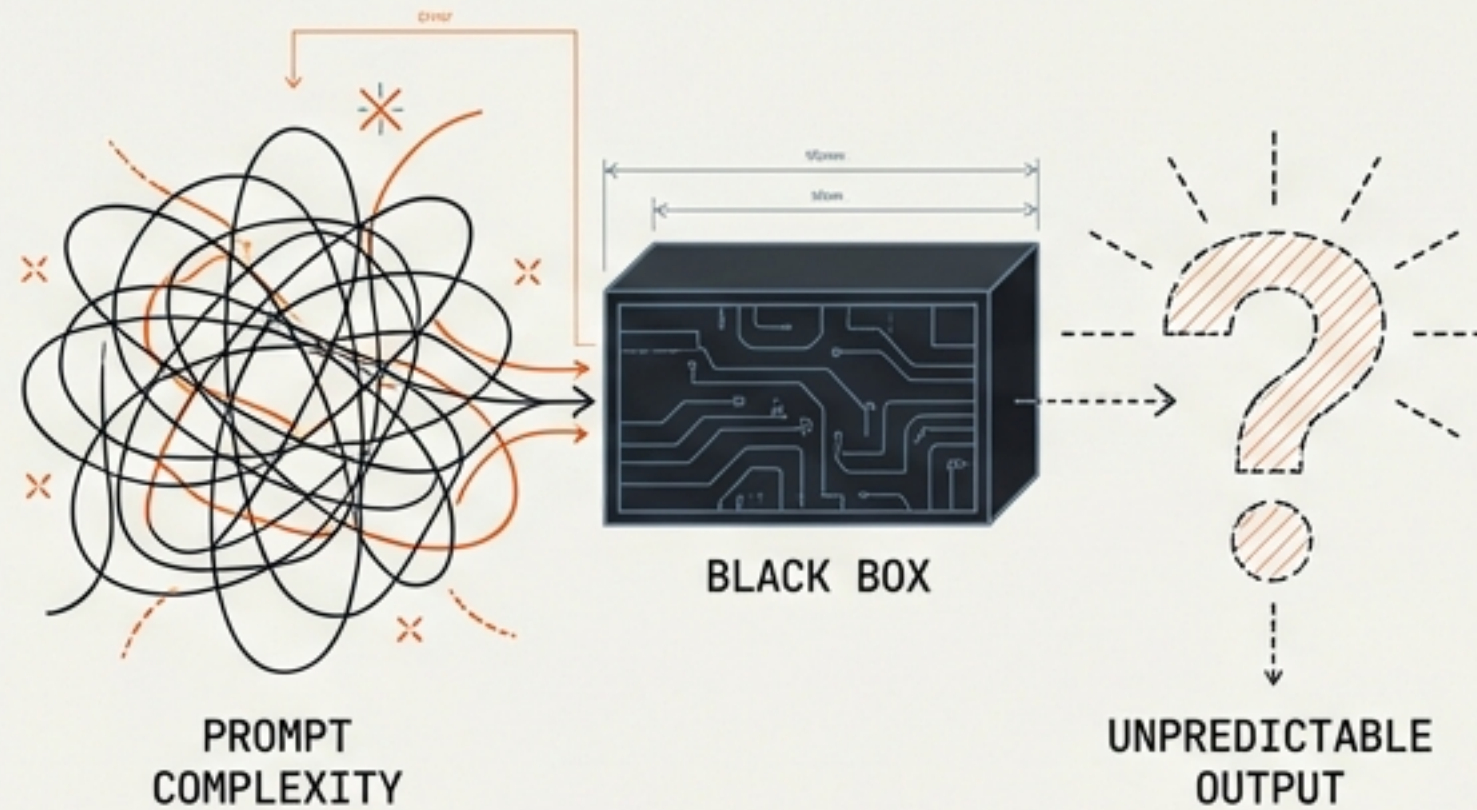
Moving from 'Chatting with Bots' to Structured Software Engineering.



Based on the workflow of expert developers using Anthropic's Claude and advanced LLMs.

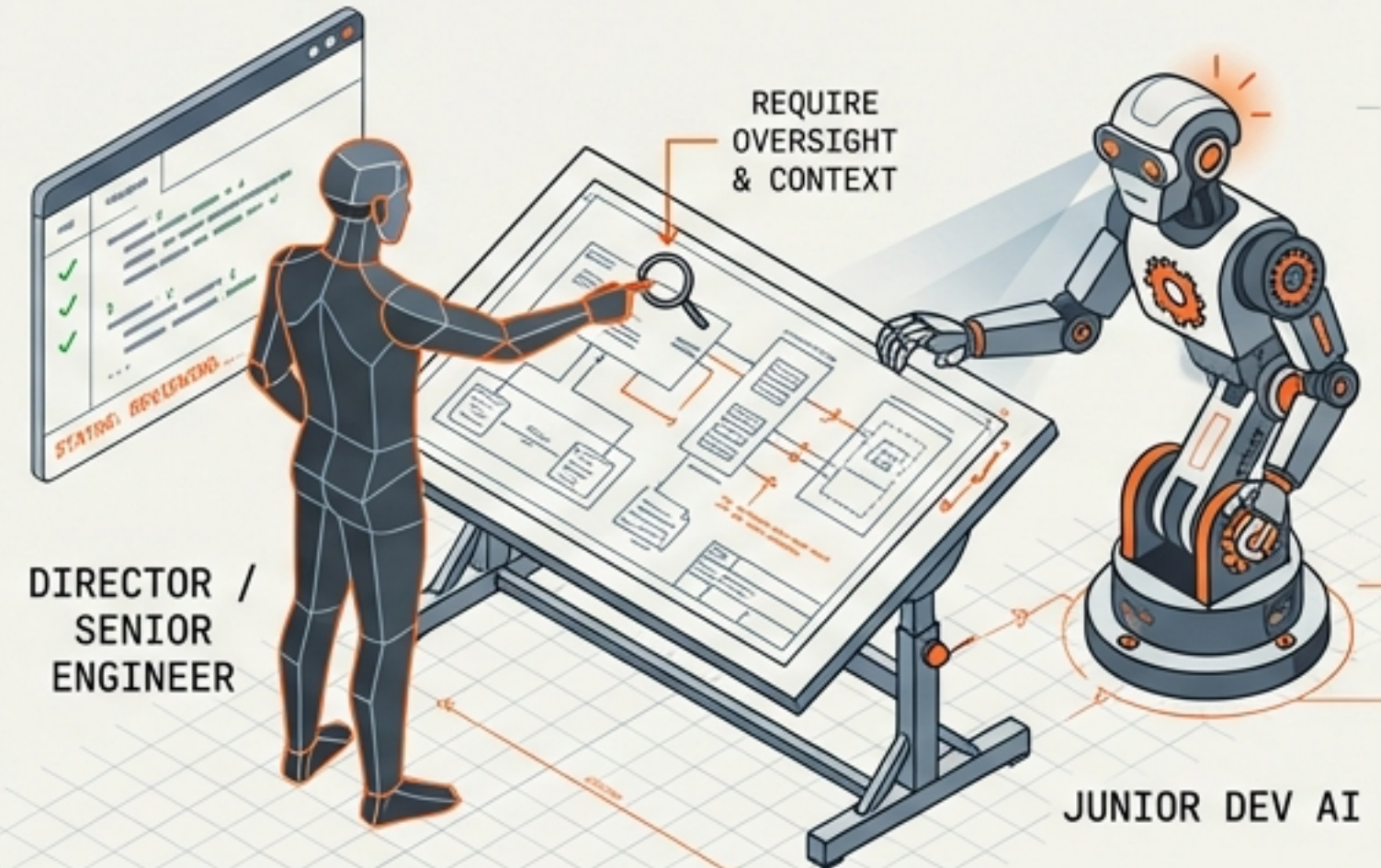
You are the Director, not just the Doer

THE TRAP: The Oracle Model



Treating AI as a magic box. Blind handoffs lead to chaos, hallucinations, and "lost in the middle" failures.

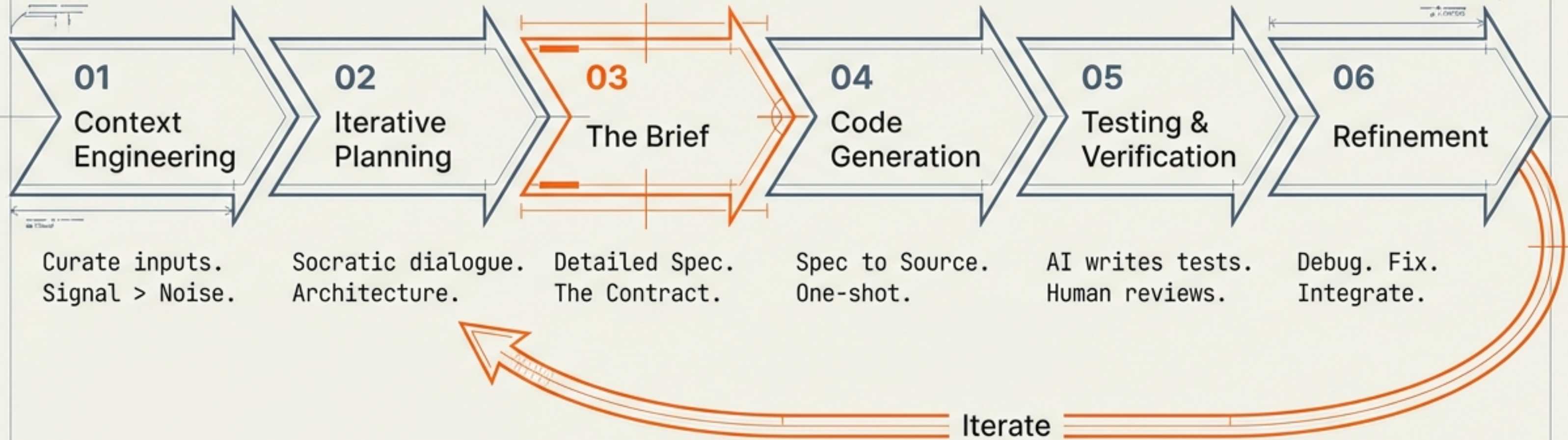
THE SOLUTION: The Junior Dev Model



Treating AI as a talented but over-confident junior engineer. Fast and knowledgeable, but requires strict oversight, context, and judgement.

「The AI augments my capabilities but
I remain the director of the show.」

The Six-Stage Augmented Cycle



A structured loop combining classic engineering with Jan 2026 LLM capabilities.

Context Engineering: Signal over Noise

Phase 1

THE WRONG WAY



Lost in the Middle Phenomenon

Large, unstructured inputs exceed window capacity, causing hallucination.

THE RIGHT WAY



- Don't dump the repo. LLMs degrade with noise.
- Use tools to create a **single, high-signal** text file.
- Include **only relevant** interfaces and data models.
- **Goal:** Avoid confused output.

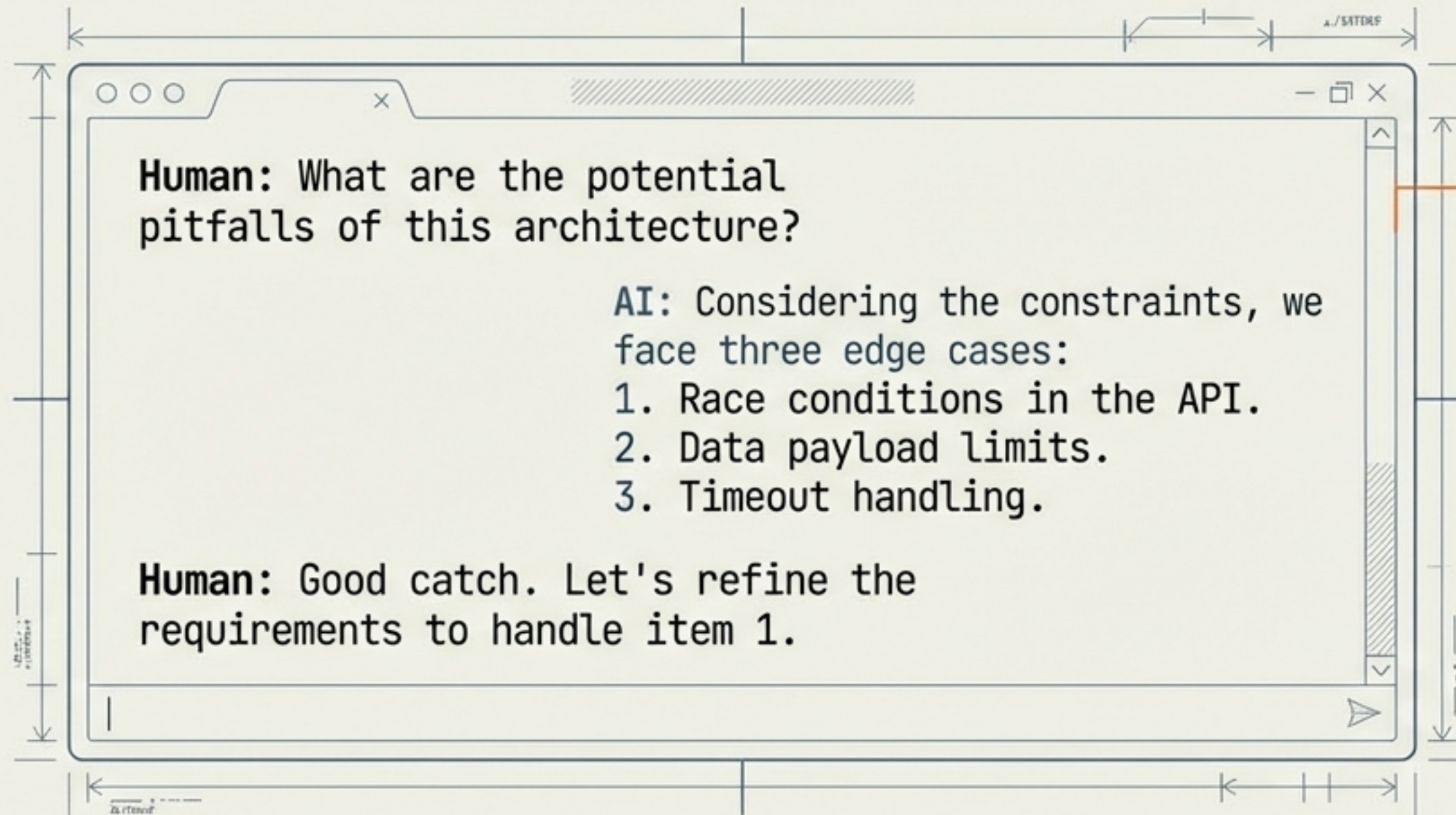
High-Signal Curated Input

Filtered, relevant data stream maximizes LLM performance.

“Context engineering isn’t about stuffing more into the prompt — it’s about curating smarter.”

Iterative Planning & Socratic Dialogue

Phase 2

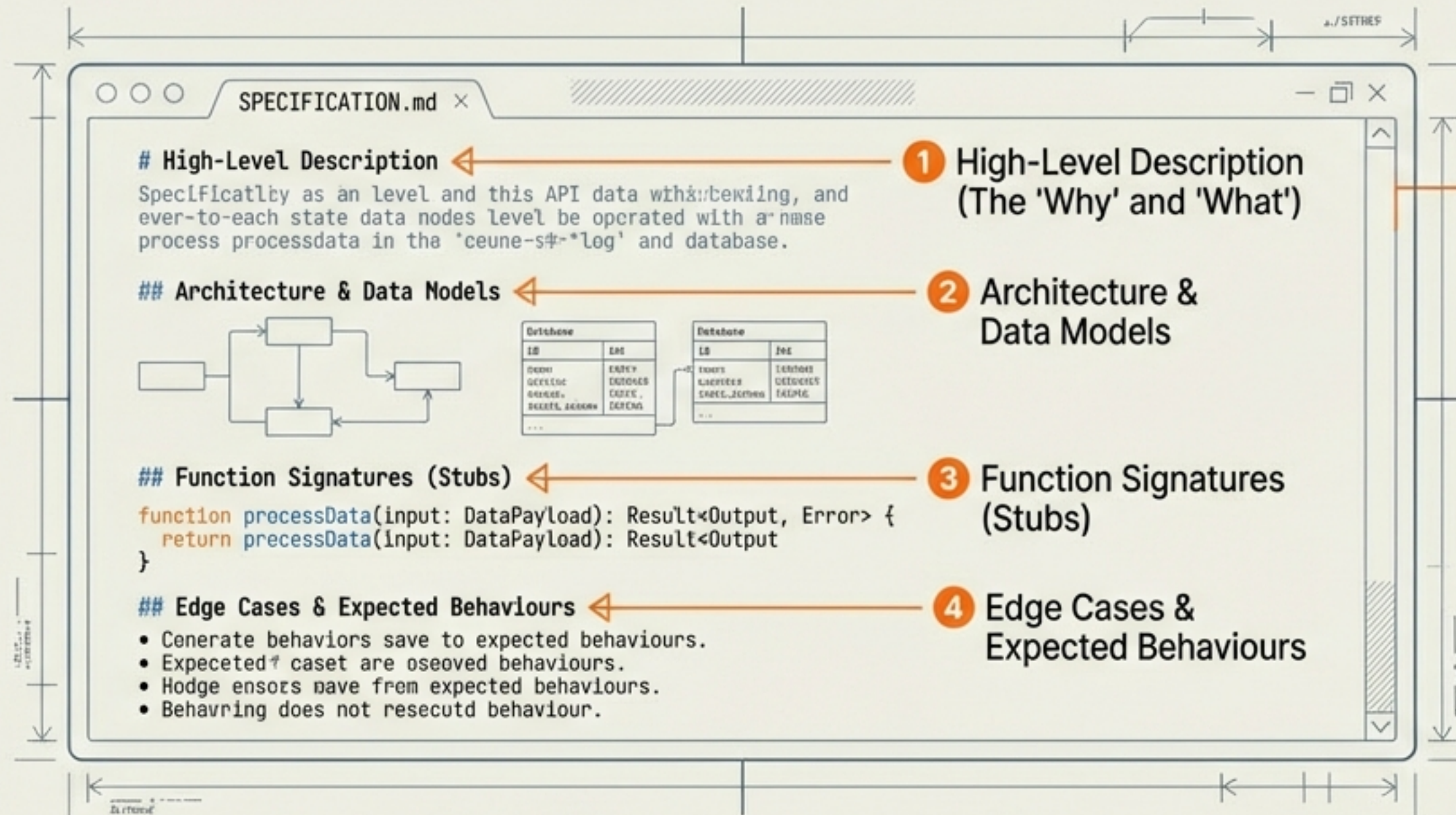


The Rule:
Code is the last step.
Thinking is the first.

- Engage in a back-and-forth design session.
- Ask the LLM to challenge your assumptions.
- Perform 'Waterfall in 15 minutes' to prevent wasted cycles.

The Brief: The Cornerstone

Phase 3

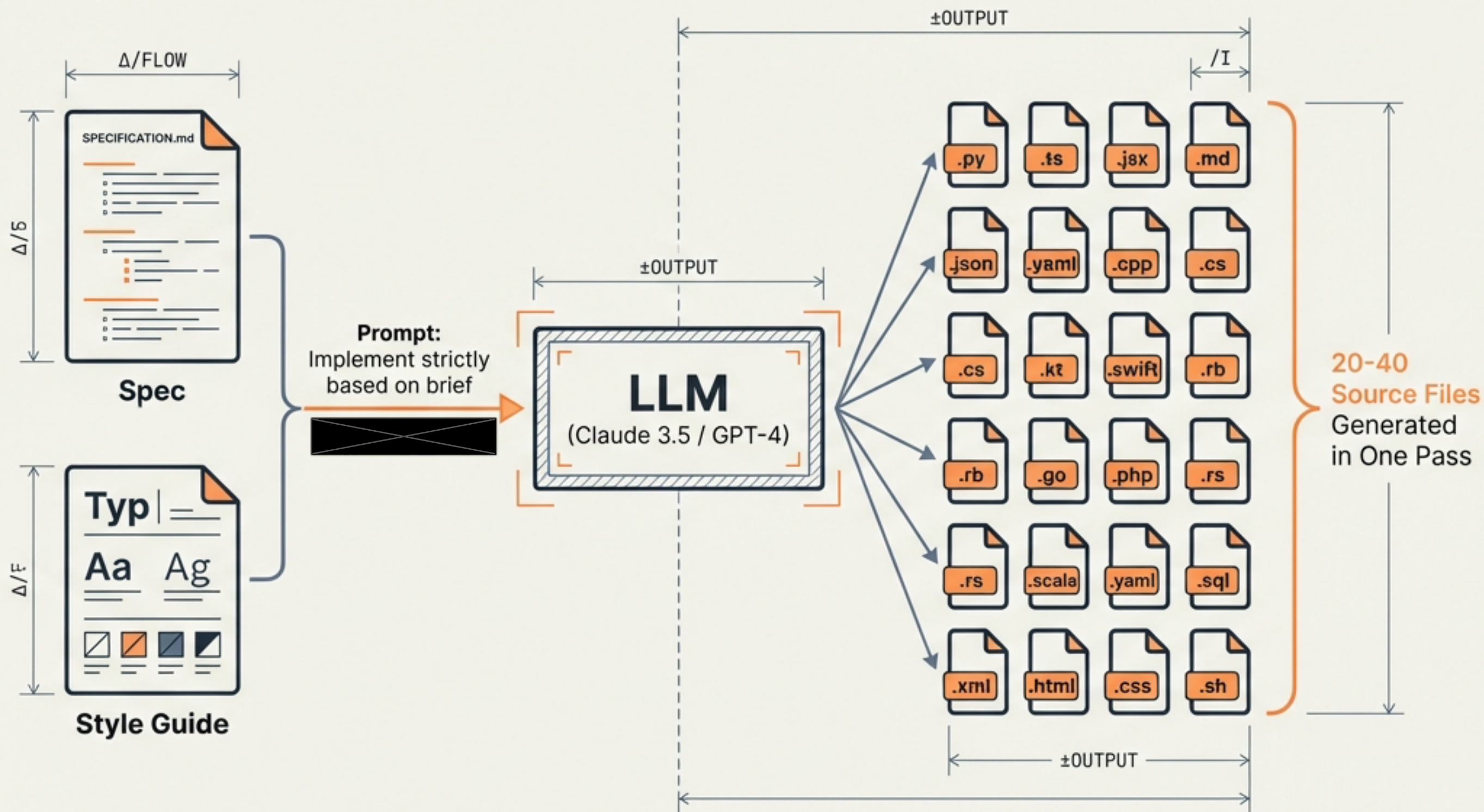


Inter
The Spec is
the Contract.

- This 1000+ line document creates a Single Source of Truth.
- It allows you to start a fresh context window for coding, preventing conversation drift.

Code Generation: From Spec to Source

Phase 4



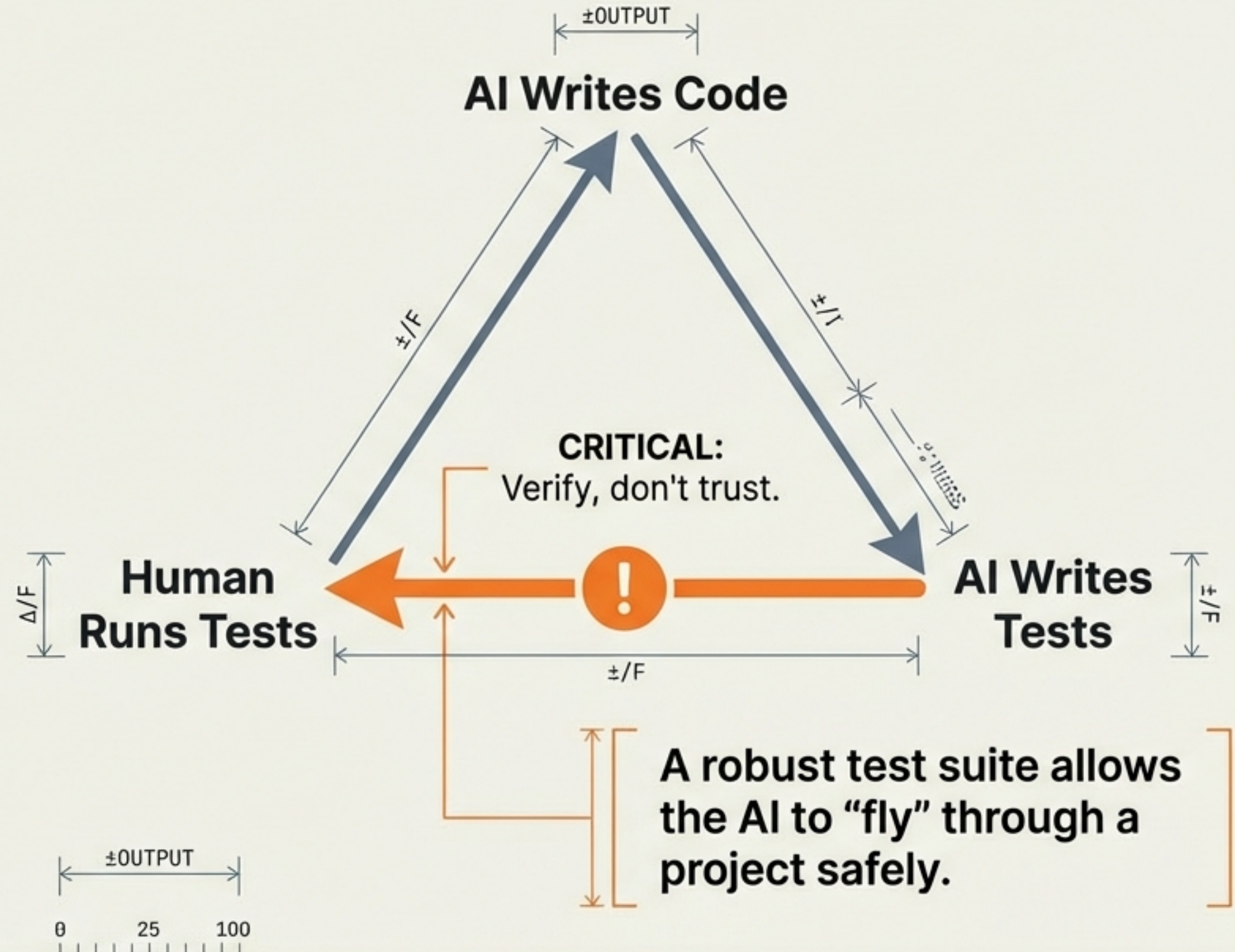
- **Strategy:** One-shot generation vs Chunking.
- With a solid Spec, modern models can generate massive volume while maintaining coherence.
- **Tip:** Use context-heavy models for depth, reasoning models for critique.
- **Goal:** Get 90% of the work done instantly.

Testing is a First-Class Citizen

Phase 5

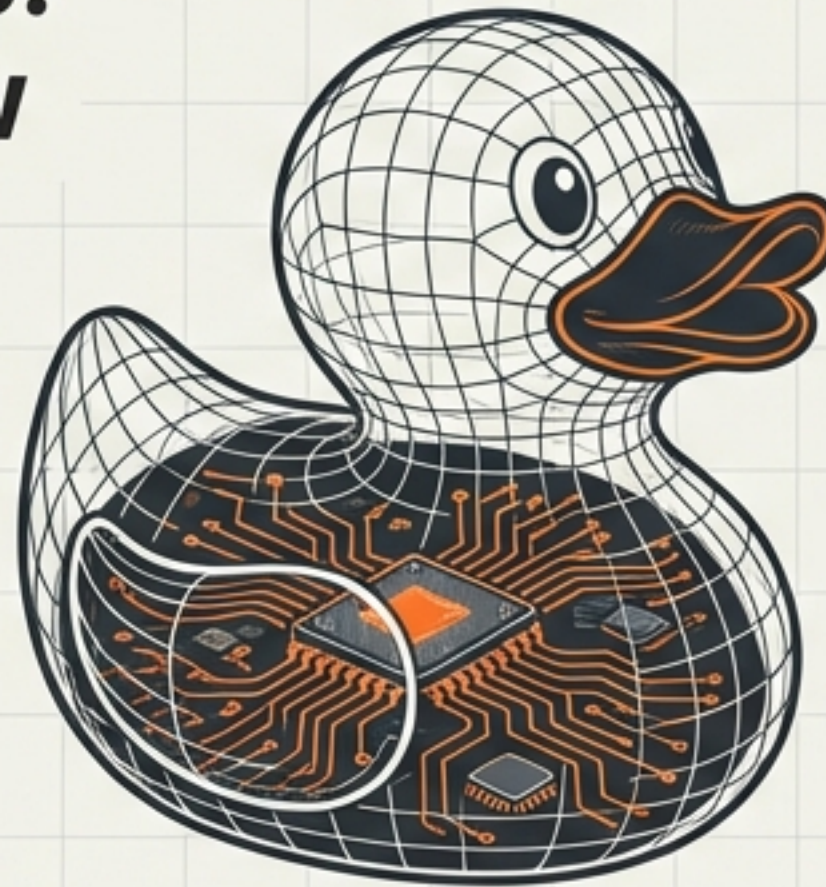
Trust, but Verify (Rigorously).

1. Prompt the LLM to write unit tests for the code it just generated.
2. The tests serve as documentation for how the code **should** work.
3. If the AI misses logic in the code, it often catches it when forced to write the test.

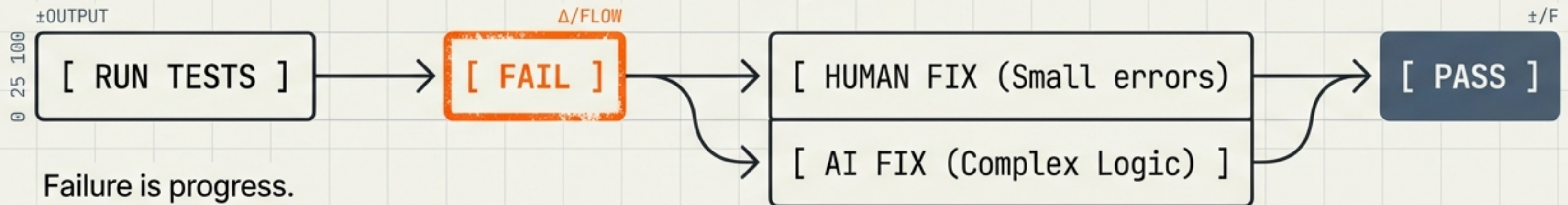


The Refinement Loop: Debugging as Review

Phase 6



The AI
Rubber Duck



Failure is progress.

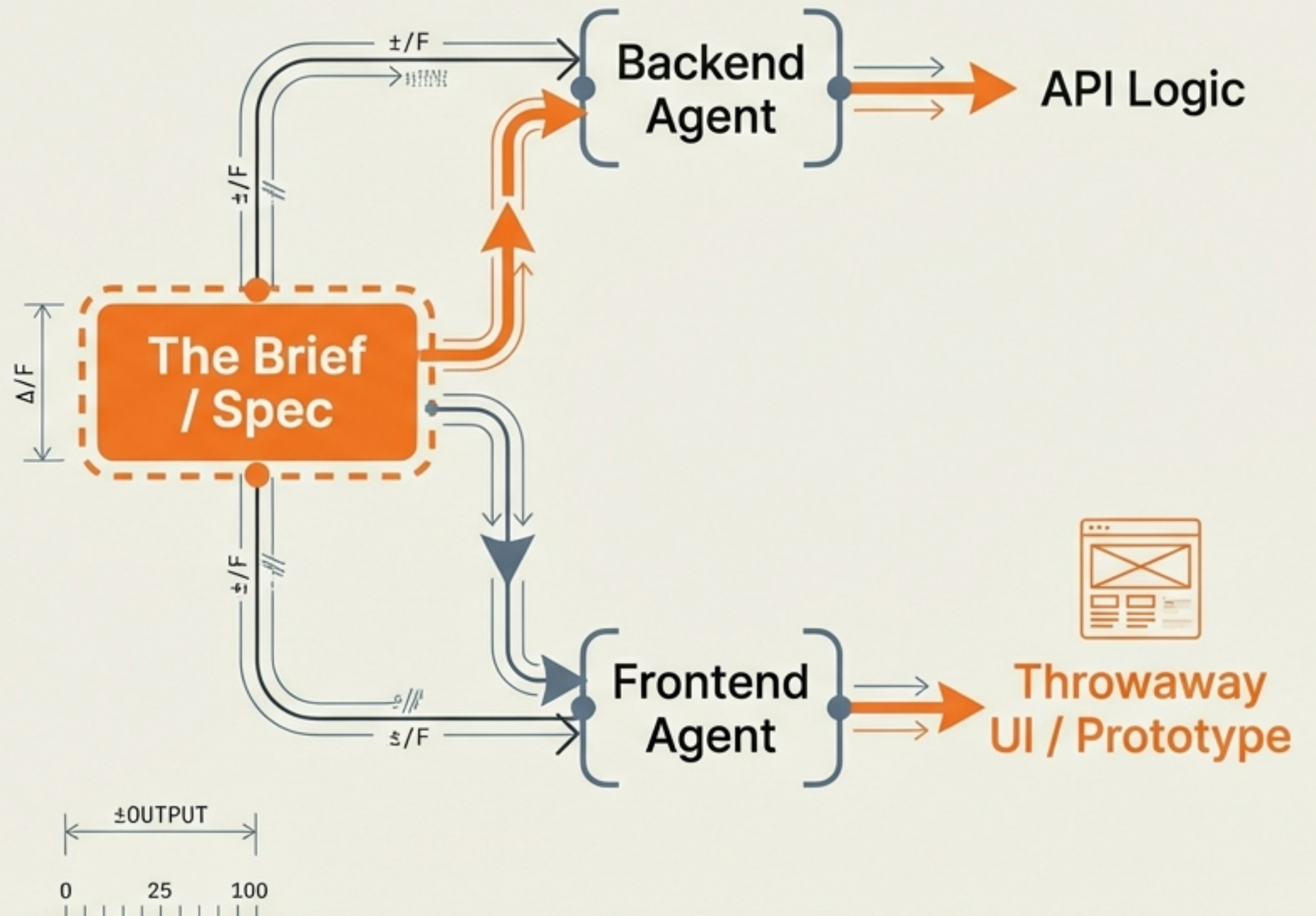
Debugging is the primary method for the human to understand the AI-generated codebase.

Reviewing code via debugging ensures you own the output.

Extended Applications: UI & Prototyping

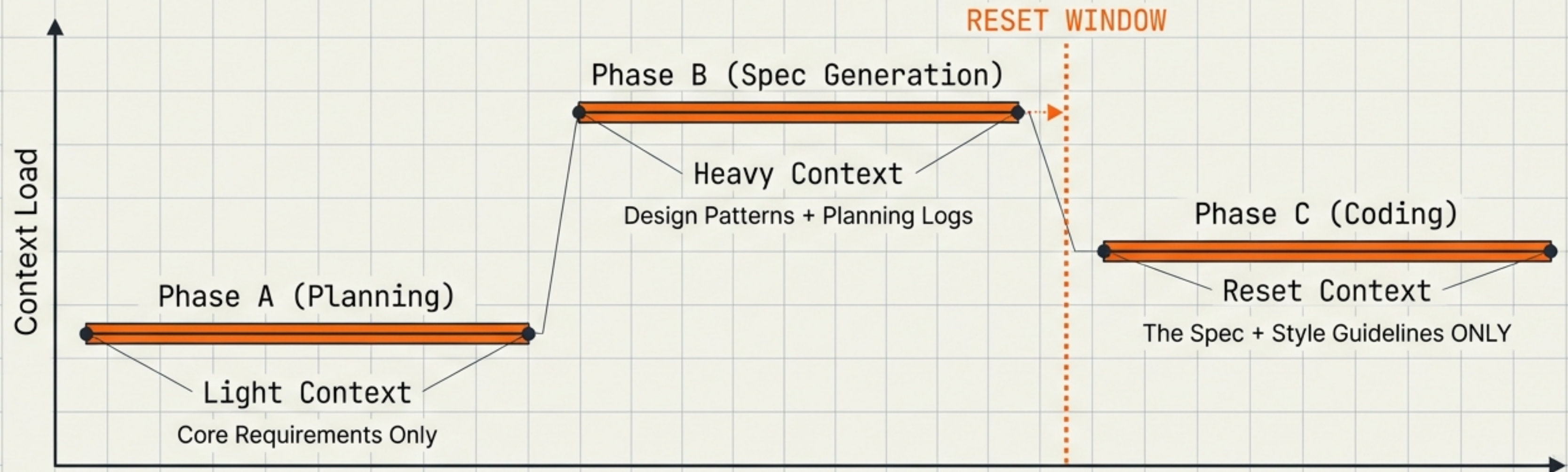
Concept: Iterative Flow Development (IFD)

1. The workflow applies beyond backend logic.
2. Use IFD to spin up throwaway frontends to visually test complex backend APIs.
3. Multi-Agent Workflow: Orchestrate a pipeline where one AI designs, one codes, and another documents.



The Mechanics of Context Phasing

Managing the Context Window

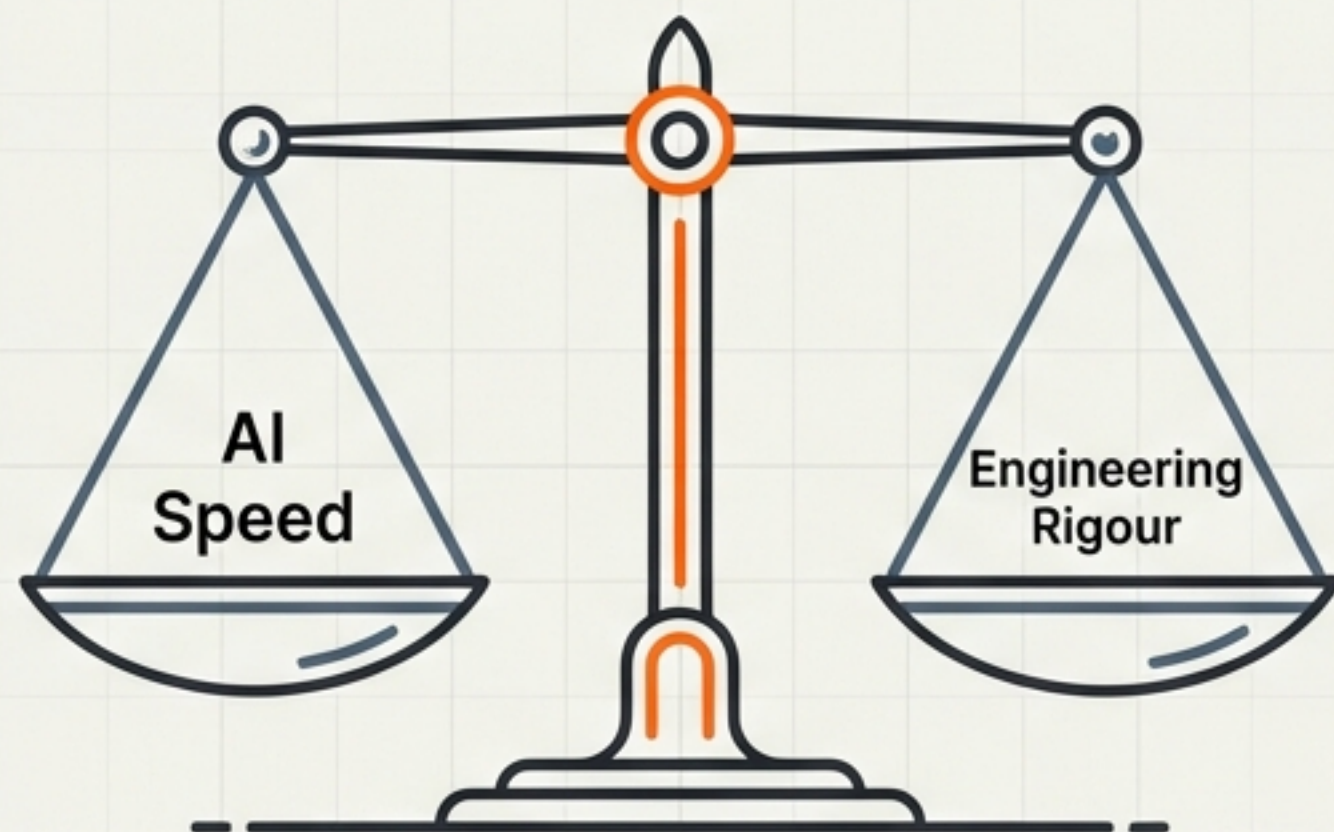


Delivering information in layers prevents token waste and confusion.

Resetting the window at critical junctions guarantees the model focuses only on the approved plan, avoiding the messy brainstorming history.

The Discipline Paradox

Engineering discipline is *more* important, not less, when using AI.



- ✓ **Curate Context** (Don't dump).
- ✓ **Review Architecture** (Don't rubber stamp).
- ✓ **Read Tests** (Don't assume coverage).
- ✓ **Maintain Standards** (Style guides are essential).

“AI doesn’t replace engineering discipline; it amplifies it.”

Owning the Code

- 1. Start with a plan.
- 2. Maintain oversight.
- 3. Be proudly accountable.

